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NPTEL ONLINE CERTIFICATION COURSE

SUSTAINABLE TRANSPORTATION SYSTEMS

Lecture 40: Modelling of Transport Emissions-I

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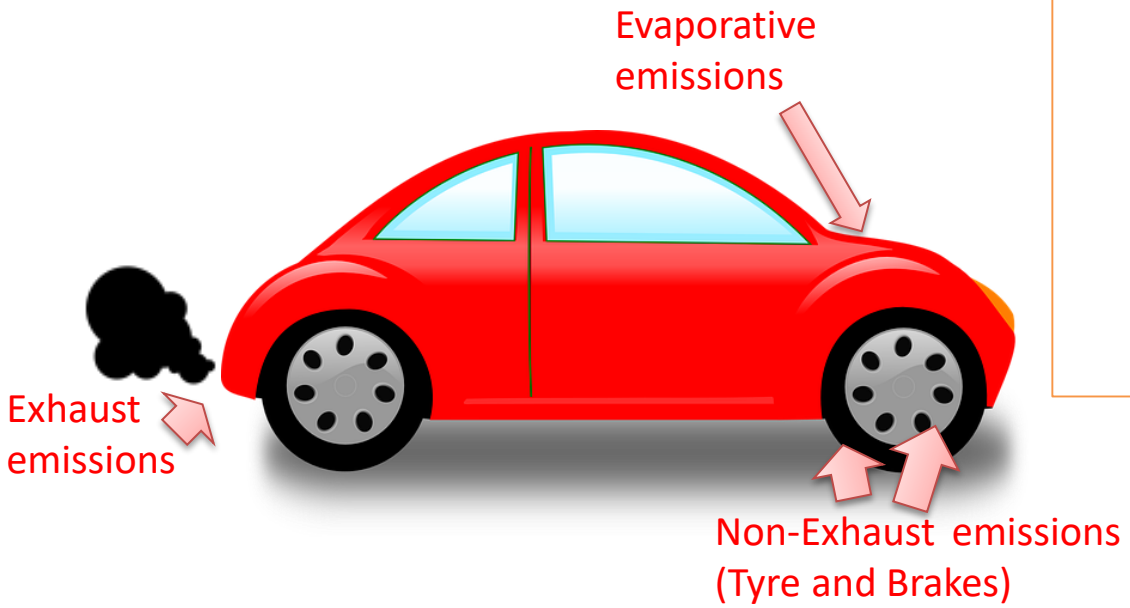
Dept. of Civil Engineering



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Vehicular Emissions



- Three types of Vehicular emissions:
  - Exhaust emissions
  - Evaporative emissions
  - Non Exhaust emissions (Tyre and Brake)

Source: (Ajay Singh Nagpure, 2011)

Types of Vehicular Emissions

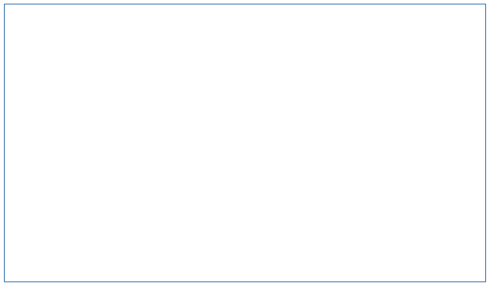
- Exhaust emissions
  - Carbon monoxide (CO)
  - Hydrocarbons (HC)
  - Nitrogen oxides (NO_X)
  - Carbon dioxide (CO_2)
  - Particulate matter (PM)
  - Benzene
  - 1-3 Butadiene
  - Formaldehyde
  - Polycyclic Aromatic Hydrocarbons (PAH)
  - Acetaldehyde
  - Total Aldehyde
  - GHG Emissions

Source: (Ajay Singh Nagpure, 2011)

Types of Vehicular Emissions (cont'd..)

- Evaporative emissions
  - Volatile Organic Compounds (VOCs) / Hydrocarbons

- Non Exhaust emissions
  - Includes emissions from tyres and brakes
    - $PM_{2.5}$ and PM_{10}
    - Heavy metals



Source: (Ajay Singh Nagpure, 2011)

Emission Inventory and Vehicular Emission models

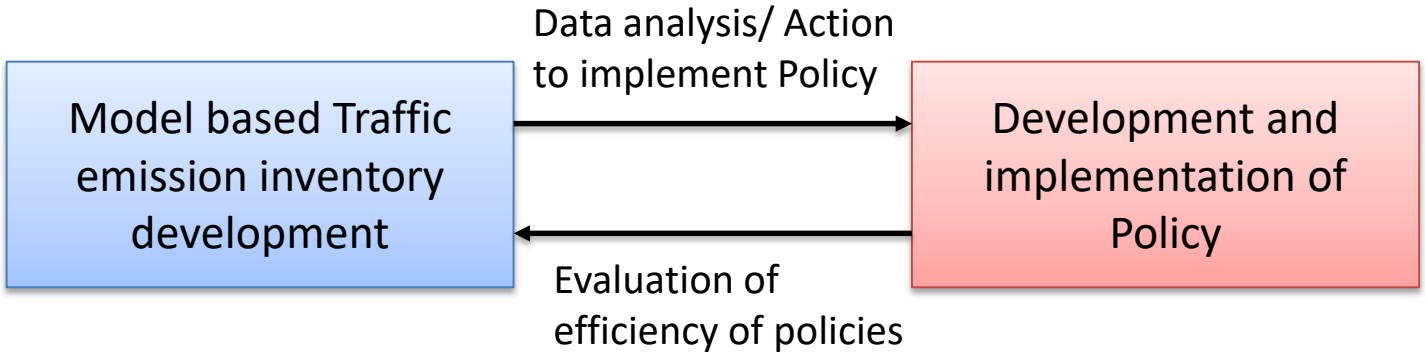
- An important tool for assessing the vehicular emissions based on emission factors, average speed, fuel consumption and the amount of traffic on the defined road type.

- Developed manually or using computer-based vehicular emission models.

- Vehicular emission models are developed based on factors such as:
  - Vehicle type
  - Fuel category & engine technology
  - Vehicle kilometers travelled
  - Emission standards
  - Various correction factors

Source: (Ajay Singh Nagpure, 2011)

Emission Inventory and Emission Model

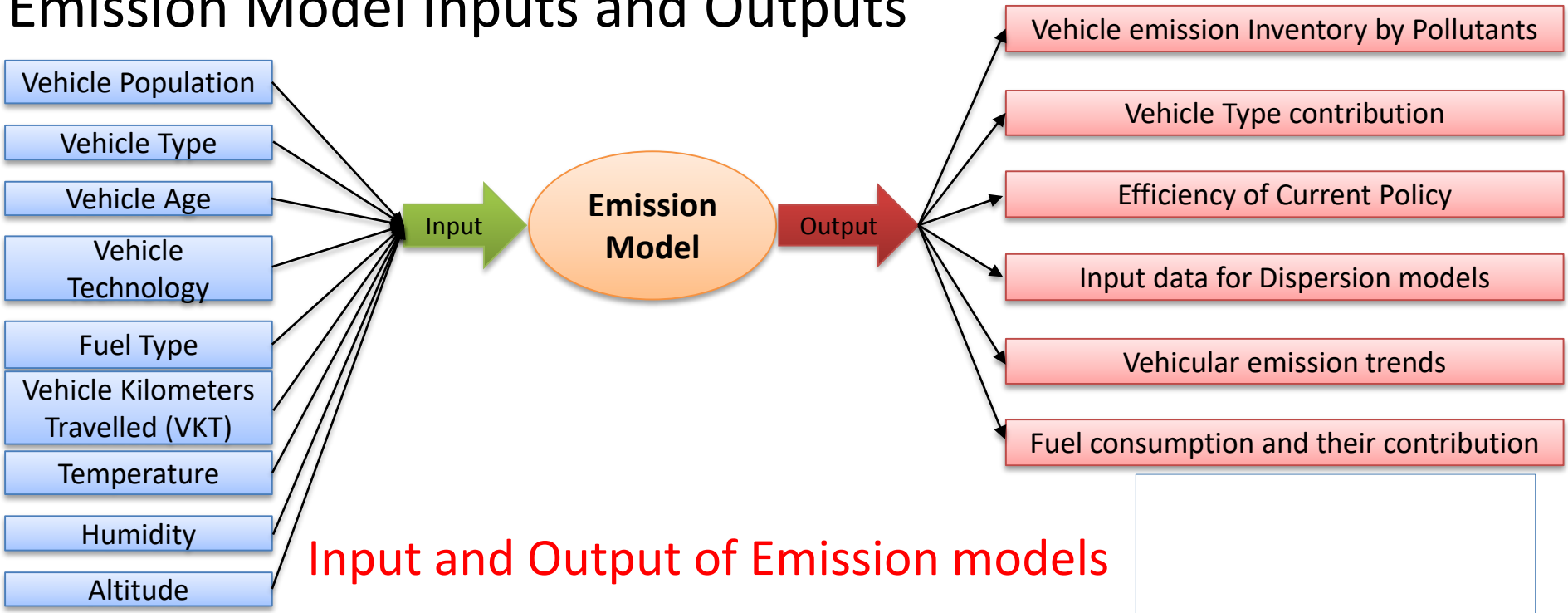


Decision Support Systems (DSS) framework for Policy development and Emission Control

Emission model, estimating emissions from road traffic is the summation of the product of emissions from all vehicular emission sources, vehicular activities and other climatic and topographic variables

Source: (Ajay Singh Nagpure, 2011)

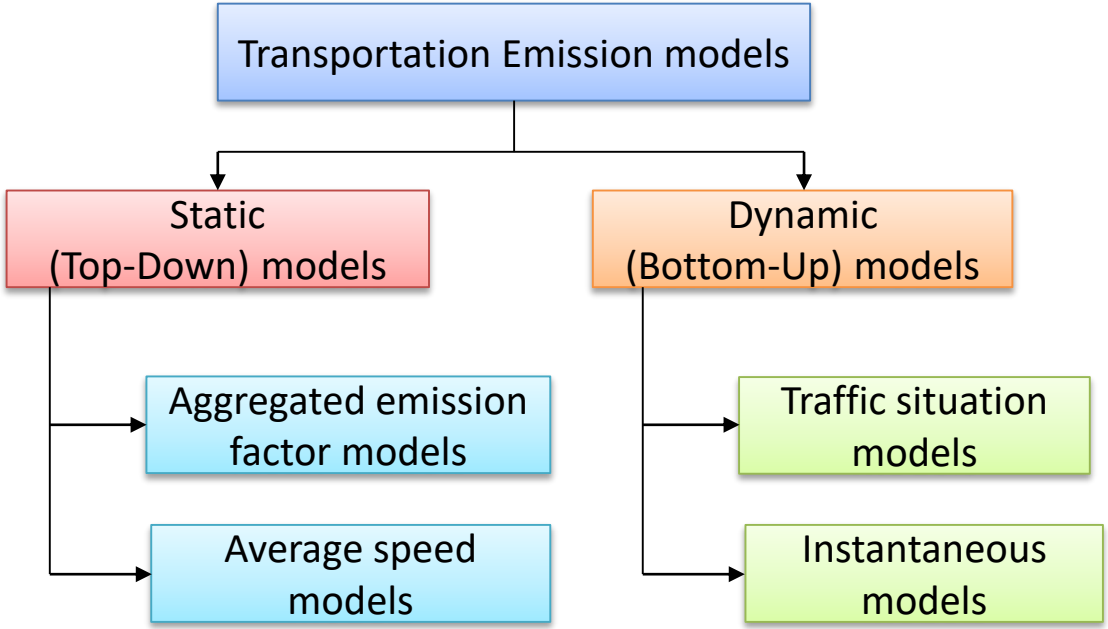
Emission Model Inputs and Outputs



Input and Output of Emission models

Source: (Ajay Singh Nagpure, 2011)

Classification of Transportation Emission models



Top-Down models or Macro-scale models works on macro scale only, whereas Bottom-Up models or Micro-scale models works on detailed and micro-level data.

Schematic diagram of the classification of the Transportation emission models

Source: (Ahmed Elkafoury et. al., 2013)

Comparison between Dynamic and Static emission models

	Dynamic models		Static models	
	Traffic situation models	**Instantaneous models**	**Average speed models**	**Aggregate emission factor models**
Base of modeling emission factors	Speed and traffic factors	Engine power	Average speed of trip	VKT, Transport productivity or Fuel consumption
Modeling scale	Micro-scale	Micro-scale	Macro-scale	Macro-scale
Usage	Street level	Street level	National or Strategic emission estimation	Emissions of unregulated pollutants
Limitations	Need detailed statistics about speed and traffic situations	Needs detailed and expensive data	Neglects cycles and driving behaviours of same average speeds	Cannot be used for situations, not explicitly defined

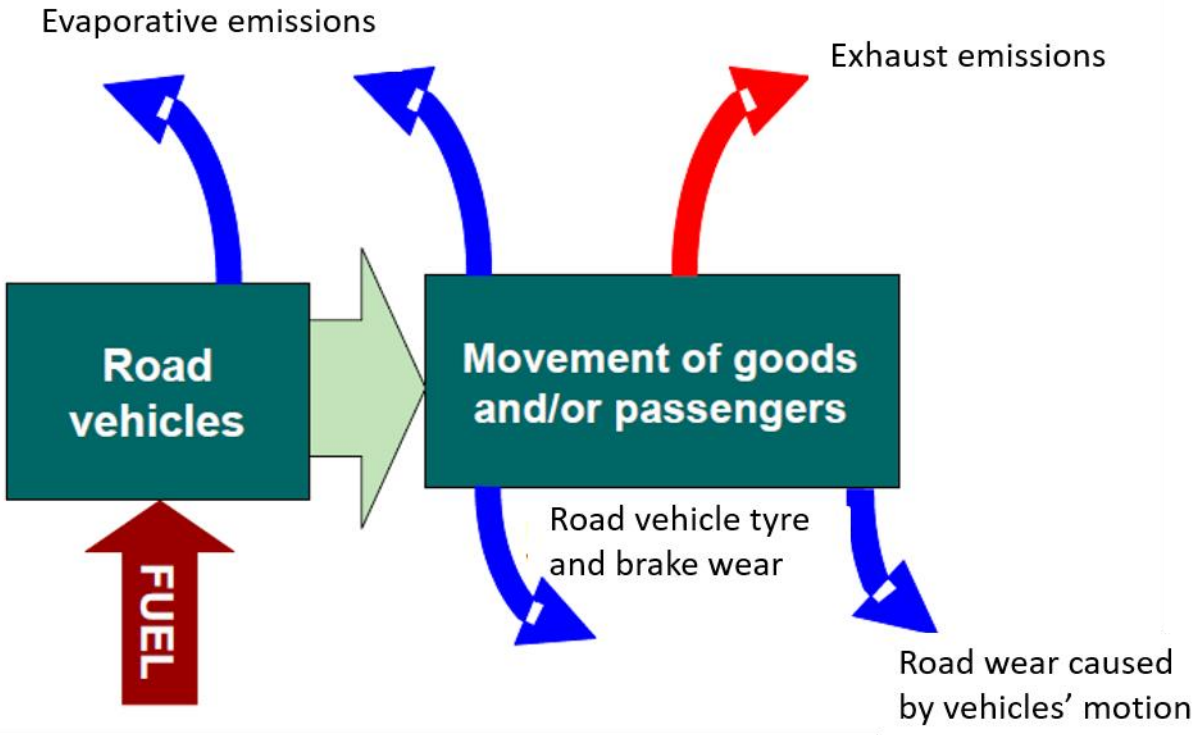
Source: (Ahmed Elkafoury et. Al, 2013)

Examples of some Vehicular Emission models

- MOBILE model series, USEPA
- MOtor Vehicle Emission Simulator (MOVES), USEPA
- EMission FACtor (EMFAC) model, California Air Resources Board (CARB)
- COmputer Program to calculate Emission from Road Traffic (COPERT), European Environment Agency (EEA)
- Queensland Government Environmental Protection Agency (QGEPA model), Australia
- International Vehicle Emissions (IVE) model, funded by the USEPA

Source: (Ajay Singh Nagpure, 2011)

What is estimated or predicted using models?



Source: (European Environment Agency, 2020)

MOBILE, USEPA

- MOBILE model series, was developed by the USEPA, for calculating emissions from highway vehicles in the U.S., except California.
- MOBILE6.2 was the last version of the series, in 2004. (Later updated to MOVES series)
- Calculates emissions of HC, NOx and CO from cars, motor vehicles, light and heavy duty vehicles.
- Superseded by the MOtor Vehicle Emission Simulator (MOVES).

Source: (Ajay Singh Nagpure, 2011)

MOBILE, USEPA (cont'd..)

- MOBILE estimates emission based on external geographical and climatic factors such as:
  - Ambient temperature
  - Humidity
  - Altitude

- Calculates basic emission rates for an average speed of 31.5 km/h for light duty gasoline vehicles, classified in terms of model year and vehicle technology.

Source: (Ajay Singh Nagpure, 2011)

MOBILE, USEPA (cont'd..)

- MOBILE computes the hot running emissions of per unit distance (e_x) in freeway and arterials as a function of travel speed for 5 mile/hr speed increments.

$$e_x = SCF\ (BER + EO)$$

Where,

e_x = emission in per unit distance

SCF = Speed Correction factor

BER = Basic emission rate

EO = emission offset, a correction factor to account for BER "off-cycle" emissions, i.e. additional emissions due to high power operation, not included in the BER.

Source: (Ajay Singh Nagpure, 2011)

MOtor Vehicle Emission Simulator (MOVES)

- MOtor Vehicle Emission Simulator (MOVES) is currently the state-of-the-science emission system of the USEPA.

- Estimates emissions for mobile sources across the U.S. except California, for air pollutants, GHGs and air toxics.

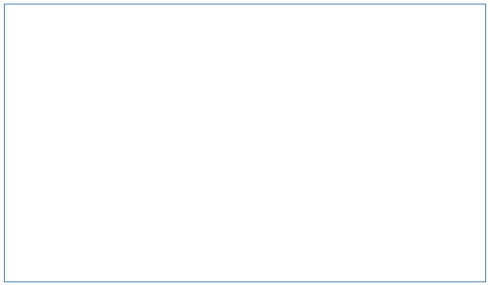
- MOVES3 is the latest version of the MOVES.
  - Follows MOVES2010 and MOVES2014 versions.

Source: (USEPA, https://www.epa.gov/moves)

MOtor Vehicle Emission Simulator (MOVES) (cont'd..)

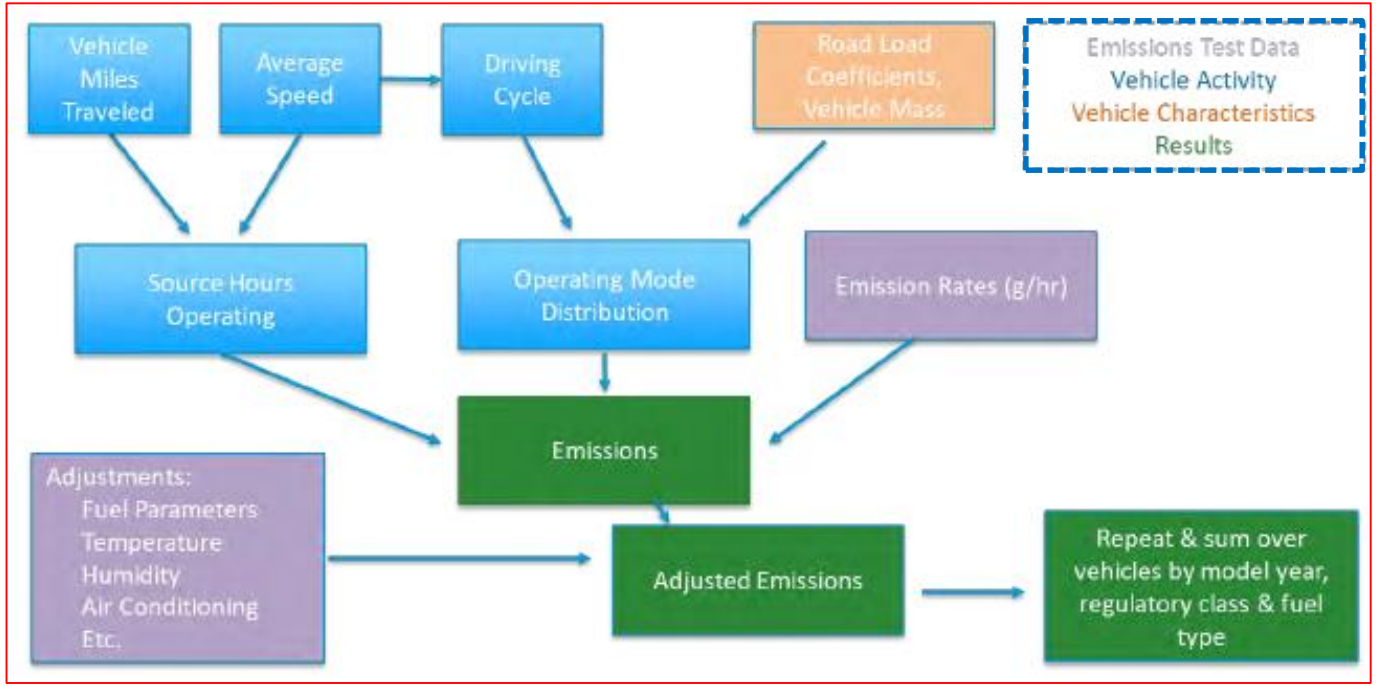
- Estimates different types of emissions such as:
  - Engine running, startling, hoteling (extended idling), evaporative, brake and tyre emissions.

- The input to MOVES model includes:
  - Default National averages such as Vehicle counts, Vehicle Miles Travelled (VMT), temperature, fuel, location and other country specific inputs.



Source: (USEPA, https://www.epa.gov/moves)

Calculating Running Emissions for Onroad vehicles in MOVES3



Source: (USEPA, 2021)

EMission FACtor (EMFAC) model

- EMission FACtor (EMFAC) model was developed by the California Air Resources Board (CARB).
- Estimates the emission inventories of onroad mobile sources in California.
- The input to the model is provided by CALIMFAC (California Motor Vehicle Emissions Factor Model), which provides emission rates and WEIGHT to estimate vehicle activity by model year.

Source: (Ajay Singh Nagpure, 2011)

EMission FACtor (EMFAC) model (cont'd..)

- The emission factors calculated from CALIMFAC are corrected in EMFAC for several correction factors.
- The Vehicle characteristics such as VKT, number of starts, number of vehicles are provided to the EMFAC by the BURDEN model to produce the emission inventory.
- Latest update is the EMFAC2021.

Source: (Ajay Singh Nagpure, 2011)

EMission FACtor (EMFAC) model (cont'd..)

- The hot running emissions in per unit distance (e_x) is calculated by the EMFAC using the equation

$$e_x = BER \times SCF = BER \times EXP(b_1(v - 27.4) + b_2(v - 27.4)^2$$

Where,

e_x = emission in per unit distance

BER = Mean emission factor value of individual vehicle (calculated as the total emissions (g) generated during laboratory test procedure using unified cycle length (mile))

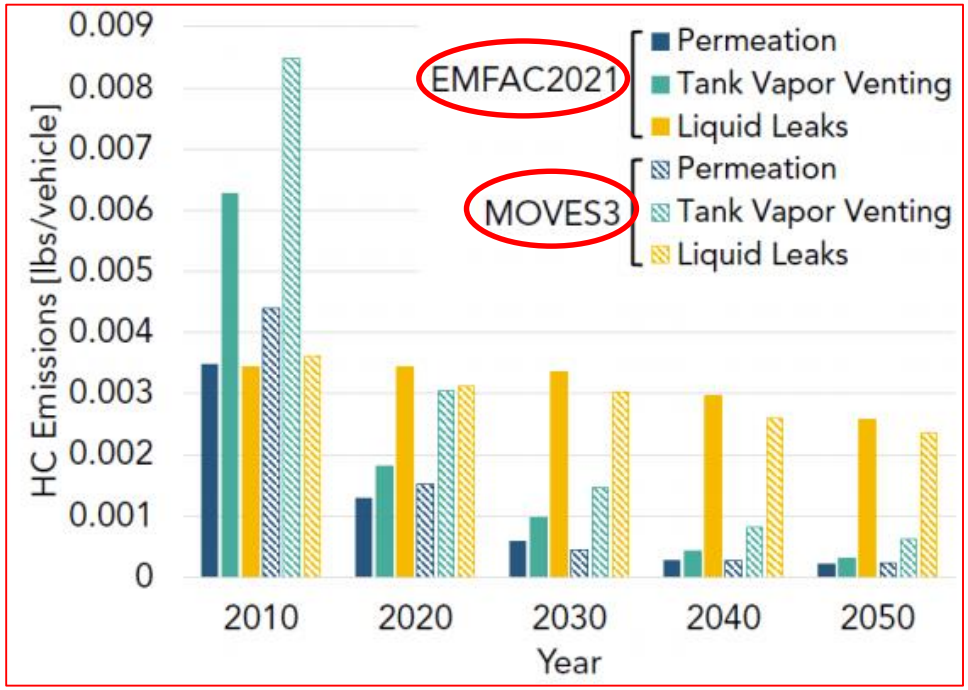
SCF = Speed Correction Factors, developed from 12 driving cycles

b_1 and b_2 = model parameters

v = speed expressed as mile per hour

Source: (Ajay Singh Nagpure, 2011)

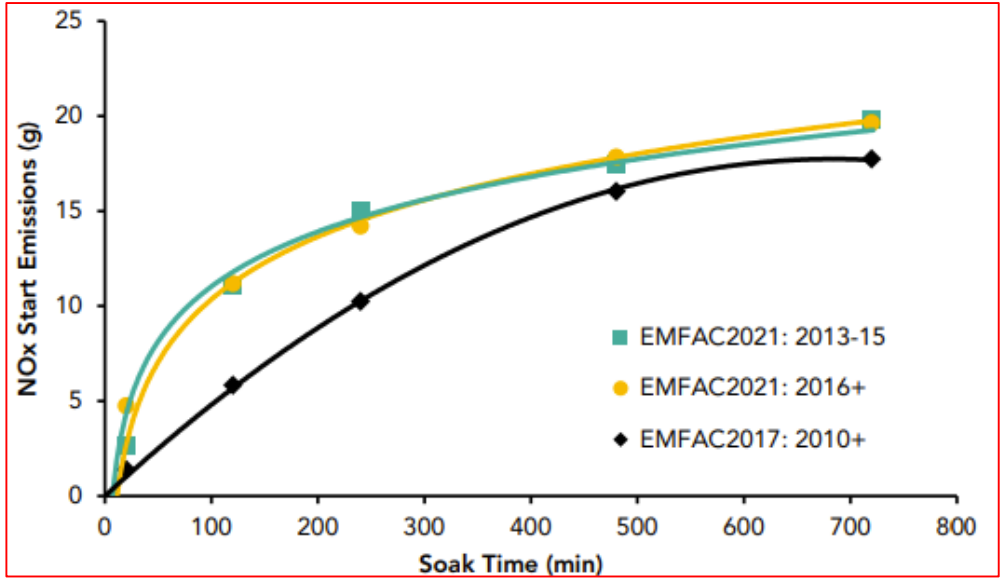
Comparison of evaporative emissions of Passenger Cars in Los Angeles county between EMFAC2021 and MOVES3



Source: (CARB, 2021)

- Show similar trends and magnitudes.
- EMFAC2021 uses California-specific inputs (meteorology, vehicle activity, and fuel information) and assumes model year distributions that are different from MOVES3.

NOx start emissions as a function of soak time for heavy-duty diesel trucks: EMFAC2021 vs EMFAC2017



- The two revised curves for EMFAC2021 both show higher NOx start emissions especially for soak times < 400 minutes.
- This indicates that NOx start emissions would be higher for EMFAC2021 if the soak times of EMFAC2017 were unchanged.

Soak time: The duration of time in which the vehicle's engine is not operating and that precedes a successful vehicle start.

Source: (CARB, 2021)

Computer Program to Calculate Emission from Road Traffic (COPERT)

- The Computer Program to Calculate Emission from Road Traffic (COPERT) is developed by the European Environment Agency (EEA).

- The Core Inventory of Air Emissions (CORINAIR) working group, the first European initiative to tackle road traffic emissions was later transformed to COPERT in 1989.

- COPERT5.4.52 is the latest update of the model released on May 2021.

Source: (EEA, https://www.eea.europa.eu/themes/air/links/guidance-and-tools/copert4-road-transport-emissions-model)

Computer Program to Calculate Emission from Road Traffic (COPERT) (cont'd..)

- COPERT computes the hot running emissions, evaporative emissions and non exhaust emissions from brakes and tyres.

- Includes 105 Vehicle categories belonging to 5 vehicle classes such as cars, light and heavy duty vehicles, urban buses and coaches and two wheelers.

- Estimates pollutants such as CO, NO_X, VOC, CH_4, CO_2, N_2O, NH_3, SO_X, diesel exhaust particulate matter (PM), PAHs, and POPs, Dioxins, Furans and heavy metals contained in the fuel such as Lead, Cadmium, Copper, Chromium, Nickel, Selenium and Zinc.

Source: (EEA, https://www.eea.europa.eu/themes/air/links/guidance-and-tools/copert4-road-transport-emissions-model)

Computer Program to Calculate Emission from Road Traffic (COPERT) (cont'd..)

- The hot running emissions in per unit distance (e_x) is calculated by the EMFAC, using the equation

$$E_{TOTAL} = EHOT + E_{COLD} + E_{EVAP}$$

$$E_{TOTAL} = EURBAN + E_{RURAL} + E_{HIGHWAY}$$

For driving conditions

Where,

E_{TOTAL} = Total emissions (g) of any pollutant

E_{HOT} = emissions (g) during stabilized (hot) engine operation

E_{COLD} = emissions (g) transient thermal engine operation (cold start)

E_{EVAP} = emissions (g) from fuel evaporation. Only relevant for NMVOC species from gasoline powered vehicles

$E_{URBAN} = ERURAL = EHIGHWAY$ = Total emissions (g) of any pollutant of respective driving situation

Source: (EEA, https://www.eea.europa.eu/themes/air/links/guidance-and-tools/copert4-road-transport-emissions-model)

What is New in COPERT 5.4 compared to its predecessor?

- Introduction of plug-in hybrid (PHEV) passenger cars
- Introduction of hybrid buses
- Revision of Euro 5 emission factors for Passenger cars (PCs) & Light Commercial Vehicles (LCVs)
- Revision of Euro 6 emission factors for PCs & LCVs
- Parameter updates – error fixes

Source: (https://www.emisia.com/wp-content/uploads/files/docs/COPERT_v5.4_Report.pdf)

Data Inputs to COPERT 5.4 for Plug-in Hybrid Electric vehicles (PHEV)

Vehicle category	Fuel type	Topology	Weight (kg)*	Drag coefficient	Battery capacity (kWh)	Engine power (kW)*	Electric motor power (kW)	Models
Small	Gasoline	Parallel	1412 (1390-1480)	0.29	27.2	28	125	BMW i3
Medium	Gasoline	Power split	1560 (1500-1625)	0.25 (0.23-0.32)	3.9 (3.2-10.9)	72.8 (63-115)	97.1 (44.5-111)	Toyota Prius
Large	Gasoline	Power split	1982 (1560-2635)	0.307 (0.25-0.36)	10.8 (5-12)	148.3 (99-410)	97 (60-130)	Simulation
Medium	Gasoline	Parallel	1672 (1500-1880)	0.292 (0.23-0.32)	9.1	102.1	102.1	VW Golf GTE
Large	Gasoline	Parallel	1982 (1560-2635)	0.307 (0.25-0.36)	10.8 (5-12)	148.3 (99-410)	97 (60-130)	Mitsubishi Outlander
Large	Diesel	Parallel	2093 (1949-2785)	0.3 (0.29-0.34)	12.5 (8-15)	164.8 (158-190)	59.3 (50-94)	Volvo V60 Twin Engine

Technology

Passenger car category

Data based on a study for Energy consumption and emission factors for PHEV passenger cars

Source: (https://www.emisia.com/wp-content/uploads/files/docs/COPERT_v5.4_Report.pdf)

Regulated Pollutant emissions from Hybrid electric vehicles during Charge Sustaining (CS) mode: Simulation Results

CS mode		Urban 29 (km/h)	Rural 56 (km/h)	Highway 110 (km/h)
NO_x (g/km)	Small gasoline	0.0058	0.0148	0.0164
	Medium gasoline	0.0058	0.0148	0.0164
	Large gasoline	0.0058	0.0148	0.0164
	Large diesel	0.4301	0.3225	0.3859
CO (g/km)	Small gasoline	0.0651	0.0303	0.0138
	Medium gasoline	0.0651	0.0303	0.0138
	Large gasoline	0.0651	0.0303	0.0138
	Large diesel	0.0617	0.0463	0.0534
HC (g/km)	Small gasoline	0.0012	0.0007	0.0010
	Medium gasoline	0.0012	0.0007	0.0010
	Large gasoline	0.0012	0.0007	0.0010
	Large diesel	0.0012	0.0009	0.0007
PM Exhaust (g/km)	Small gasoline	0.0019	0.0015	0.0018
	Medium gasoline	0.0019	0.0015	0.0018
	Large gasoline	0.0019	0.0015	0.0018
	Large diesel	0.0021	0.0014	0.0009

Source: (https://www.emisia.com/wp-content/uploads/files/docs/COPERT_v5.4_Report.pdf)

Until more experimental data become available, the following approach is proposed in COPERT5.4.52:

- Petrol PHEV (in CS mode): use equation for petrol hybrid
- Diesel PHEV (in CS mode): use equation for diesel conventional

QGEPA model, Australia

- The Queensland Government Environmental Protection Agency (QGEPA) developed the QGEPA model.

- QGEPA is a travel speed emission model and consists of a hot running emission factor database for CO, HC, PM and NO_X, expressed as g/km.

- Emission factors available for vehicle technology classes such as light duty vehicles (petrol, diesel and LPG), heavy duty vehicles (diesel, CNG, LPG) and motorcycles.

Source: (Ajay Singh Nagpure, 2011)

QGEPA model (cont'd..)

- Separate emission factors available for cold start and evaporative emissions.
- Also provides emission factors for non regulated pollutants such as hydrocarbons, N_2O, NH_3 and PAH.
- Speed correction factors are used for evaporative running loss emissions for Cold start emissions.

Source: (Ajay Singh Nagpure, 2011)

International Vehicle Emissions (IVE) model

- Developed jointly by the University of California at Riverside, College of Engineering- Center for Environmental Research and Technology (CE-CERT), Global Sustainable Systems Research (GSSR) and the International Sustainable Systems Research Center (ISSRC).

- Funded by the USEPA.

- Takes into account various technologies and conditions that exist in the developing countries such as vehicle driving patterns, vehicle specific power (VSP), Engine stress distributions etc.

Source: (Ajay Singh Nagpure, 2011)

International Vehicle Emissions (IVE) model (cont'd..)

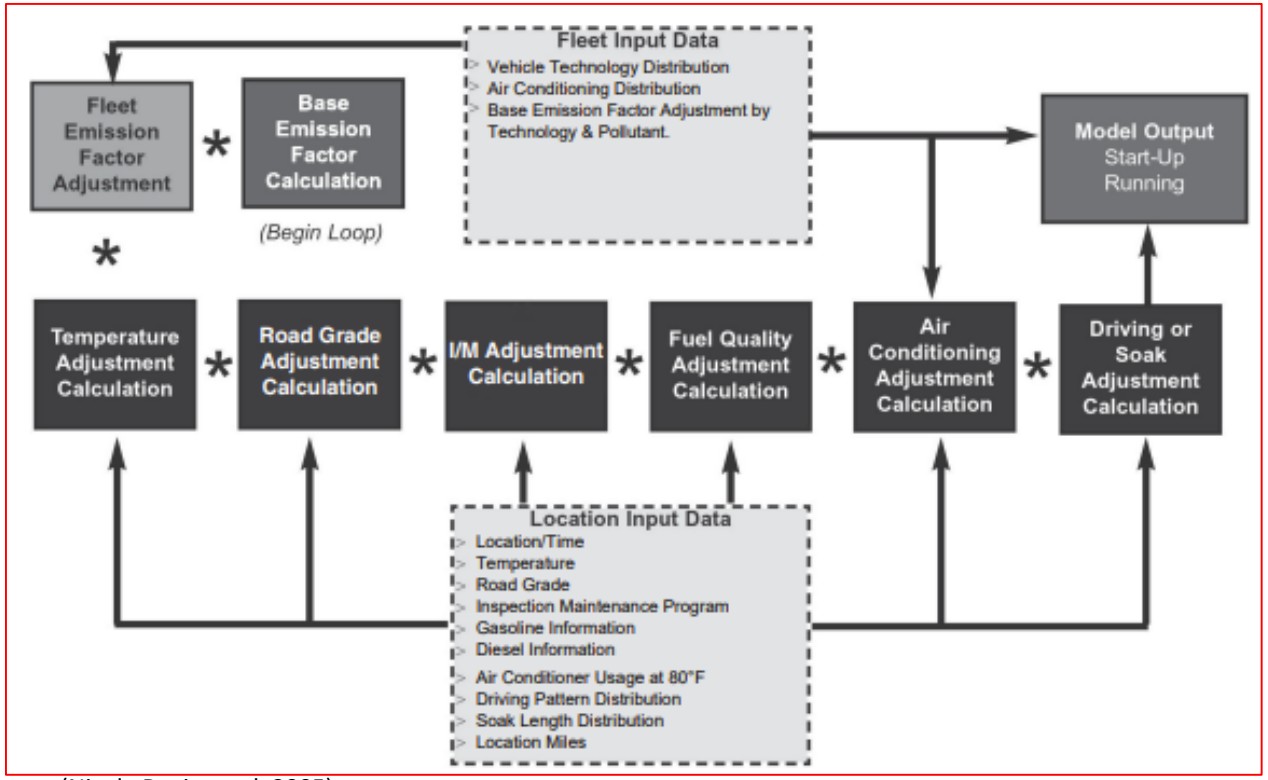
- Inputs to the model includes:
  - Type and age of vehicle fleet
  - Local conditions such as ambient temperature and pressure
  - Fuel specifications

- Estimates Criteria pollutants, GHGs and toxic pollutants.

- Pollutants includes CO, VOC, NO_X, $PM_{2.5}$, PM_{10}, CO_2, N_20, CH_4, NH_3, Benzene, Lead, 1,3 Butadiene and Aldehydes.

Source: (Ajay Singh Nagpure, 2011)

Components of the IVE Model



Source: (Nicole Davis et. al, 2005)

Types of Technologies included in the IVE Model

Light-Duty Gasoline Vehicles	Light-Duty Diesel Vehicles	Light-Duty Vehicles (Ethanol, Natural Gas, Propane, Retrofits, etc.)	Heavy-Duty Gasoline Vehicles	Heavy-Duty Diesel Vehicles	Heavy-Duty Vehicles (Ethanol, Natural Gas, Propane, etc.)	Gasoline, Ethanol, Natural Gas, Propane 2- and 3-Wheelers
Carburetor	Pre-chamber injection	Carburetor/mixer	Carburetor	Pre-chamber injection	Carburetor	Carburetor
Single-pt FI	Pre-chamber injection	Single-pt FI	FI	Direct injection	FI	FI
Multi-pt FI	Direct injection	Multi-pt FI	None	FI	None	None
None	FI	None	2-way	None	2-way/EGR	Catalyst
2-Way	None	2-way	2-way/EGR	Improved	3-way	High tech
2-Way/EGR	Improved	2-way/EGR	3-way	EGR+	3-way/EGR	2-cycle
3-Way	EGR+	3-way	3-way/EGR	PM		4-cycle
3-Way/EGR	PM	3-way/EGR	EuroI	PM/NOx		
LEV	PM/NOx	ZEV	EuroII	EuroI		
ULEV	EuroI		EuroIII	EuroII		
SULEV	EuroII		EuroIV	EuroIII		
EuroI	EuroIII		EuroV	EuroIV		
EuroII	EuroIV			EuroV		
EuroIII	Hybrid			Hybrid		
EuroIV						
Hybrid						

Pt FI = petroleum-fueled fuel-injected vehicle; EGR = exhaust gas recirculation; LEV = vehicle meeting the low-emission vehicle standards; ULEV = vehicle meeting the ultra-low-emission vehicle standards; SULEV = vehicle meeting the super-ultra-low-emission vehicle standards; EuroI = vehicle meeting European Phase I standards; EuroII = vehicle meeting European Phase II standards; EuroIII = vehicle meeting European Phase III standards; EuroIV = vehicle meeting European Phase IV standards; EuroV = vehicle meeting European Phase V standards; PM = particulate matter; NOx = oxides of nitrogen; ZEV = vehicle meeting the zero-emission vehicle standards.

Source: (Nicole Davis et. al, 2005)

Vehicle and fuel types and technologies included in the IVE model

Basic Equations in IVE model

- Can estimate emissions from a single roadway or an entire area for specific time periods.

- Based on 2 basic equations:

Equation 1

$$Q_T = B_{(t)} \times K_{(base)[t]} \times K_{(temp)[t]} \times K_{(Hmd)[t]} \times K_{(IM)[t]} \times K_{(Alt)[t]} \times K_{(Cntry)[t]}$$

- **Equation 1** estimates the adjusted emission rate by multiplying the base emission rates by various Correction factors.

Source: (Ajay Singh Nagpure, 2011)

Basic Equations in IVE model (cont'd..)

$$Q_{Running} = U_{[FTP]} \times D/Uc \times \sum_t \{f_{[t]} \times Q_{[t]} \times \sum_d \{f_{[dt]} \times K_{[dt]}$$

Equation 2

- **Equation 2** weights the adjusted emission rate by the travel fraction for each technology and amount of each driving type of each technology.
- **Final step in Equation 2** is to multiply these results by the ratio of the average velocity of LA-4 driving cycle and average velocity of the modeled cycle and multiply by the distance travelled (for running emissions only).
- Result is the overall fleet running emissions for allocated time or distance (in grams).

Source: (Ajay Singh Nagpure, 2011)

The U.S. FTP-72 (Federal Test Procedure) cycle is also called Urban Dynamometer Driving Schedule (UDDS) or LA-4 cycle

Basic Equations in IVE model (cont'd..)

Where,

$B_{[t]}$ = Base emission rate in for each technology (start (g) or running (g/km))

$Q_{[t]}$ = Adjusted emission rate for LDCVs running (g)

Q = Average emission rate for entire fleet (running (g))

$f_{[t]}$ = Fraction of travel by specific technology

$f_{[dt]}$ = Fraction of each type of driving or soak by specific technology

$U_{[FTP]}$ = Average velocity from the specific driving cycle as input by user in location (kph)

$K_{(base)[t]}$ = Adjustment to the base emission rate

$K_{(Tmp)[t]}$ = Temperature Correction Factor

$K_{(Hmd)[t]}$ = Humidity Correction Factor

$K_{(Alt)[t]}$ = Altitude Correction Factor

$K_{(Fuel)[t]}$ = Fuel quality Correction Factors

$K_{(IM)[t]}$ = Inspection/Maintenance Correction Factors

$K_{(Cntry)[t]}$ = Country Correction Factors

$B_{[t]}$ = Base emission rate in for LDCVs running (g/km)

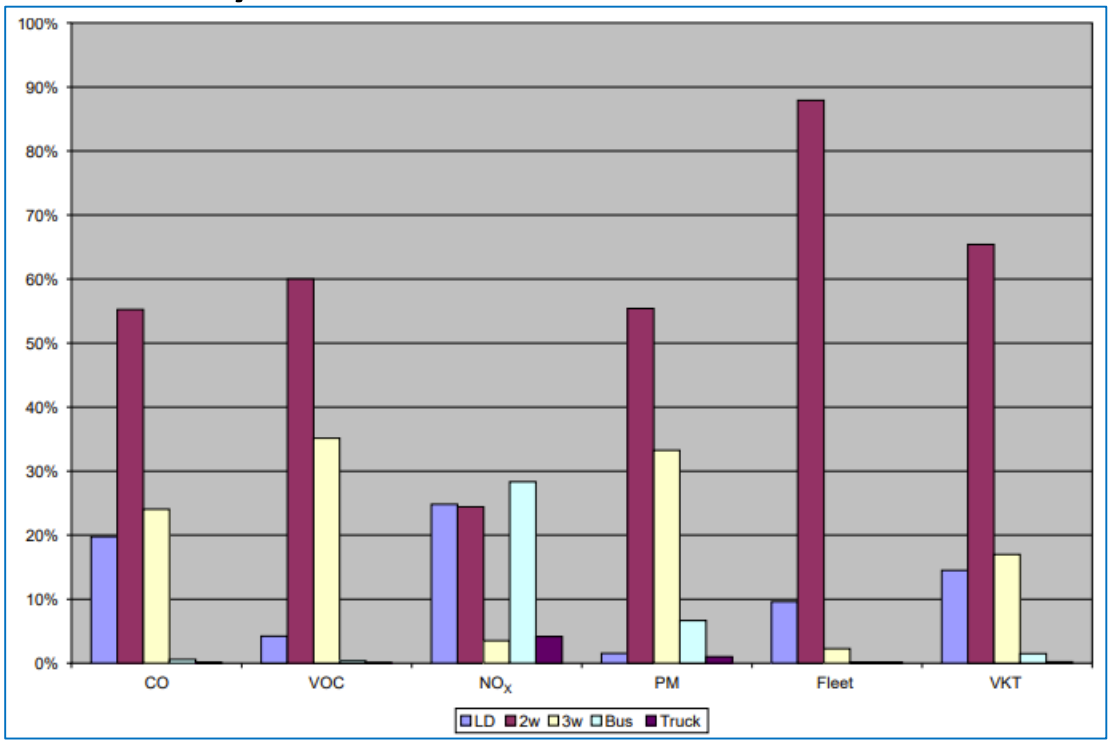
D = Distance travelled by LDCV in one day (km)

Ei = Emission of compound i (CO, NO_X, VOC)

$K_{[dt]}$ = Driving Correction Factor

Source: (Ajay Singh Nagpure, 2011)

Example of Emission contribution from each vehicle type in Pune city, India

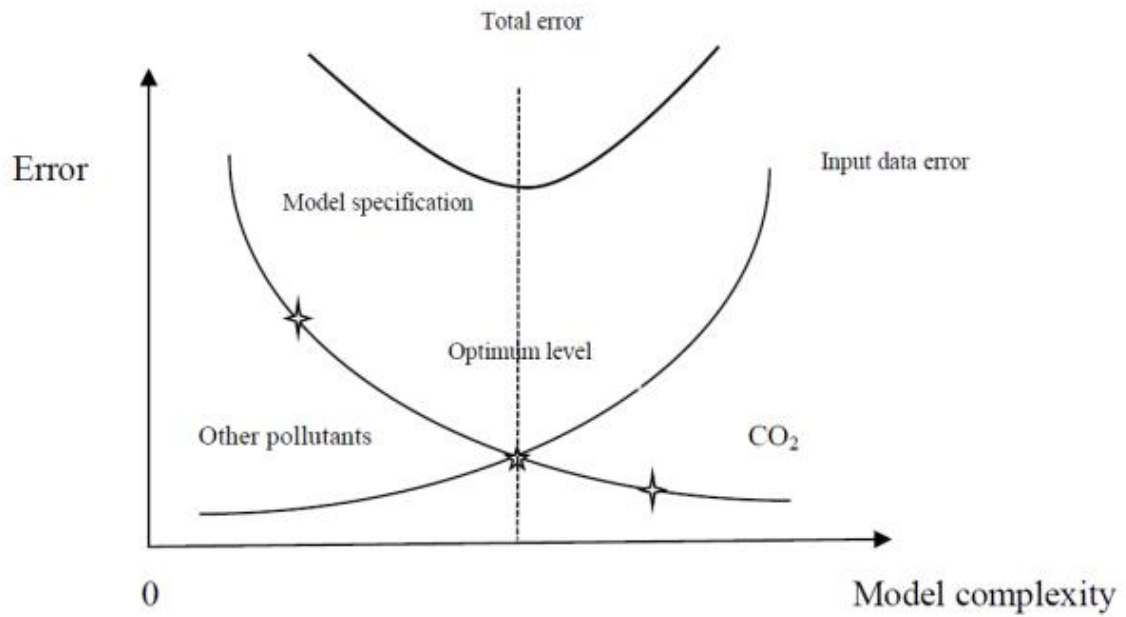


Source: (Mathew Barth et. al, 2007)

- The dominant vehicles on the roads in Pune were two-wheelers followed by three wheelers.
- Buses and Light duty four wheeler vehicles were major source of NO_X emissions.

Model Complexity and Errors in Emission models

- The reliability of emission prediction relies on two factors, namely accuracy and robustness.



- On one hand, the modelling accuracy can be improved by more complex modelling formulations.
- On the other hand, the robustness of model complexity tends to be negated by the increasing input errors.

Source: (Sicong Zhu, 2014)

Limitations of these emission models for Indian scenario

- All major models like MOBILE, MOVES, COPERT etc. are developed in developed countries such as US, Japan, Australia etc.
  - Extensive data set are available in those countries.
  - Difference in technology, infrastructure, driving cycle in comparison with Indian scenario.
  - May give erroneous results.

- Emission factor or basic emission rate of emissions for all models are US and Europe based.
  - Enhance the error in model emission outputs for use in Indian scenario.

Source: (Ajay Singh Nagpure, 2011)

(cont'd..)

- IVE model was designed for developing countries similar to India.
  - However, the complexity of the model is similar to the US and European models, which needs extensive data requirements.
  - Difficult to compile such complex datasets in Indian scenario.

- Most models are not able to give output for more than a year.
- Most models require experimental data, which are not available for Indian conditions.
- Most models are region specific.

Source: (Ajay Singh Nagpure, 2011)

The Vehicular Air Pollution Inventory (VAPI) Model by IIT Roorkee

- The VAPI model is designed specifically for developing countries like India.
- VAPI model have incorporated Climatic correction factors, making the estimations more realistic and region specific.
- The VAPI model estimations show fair agreement with the air quality observations in India.
- The model is simple to use and is available in public domain.

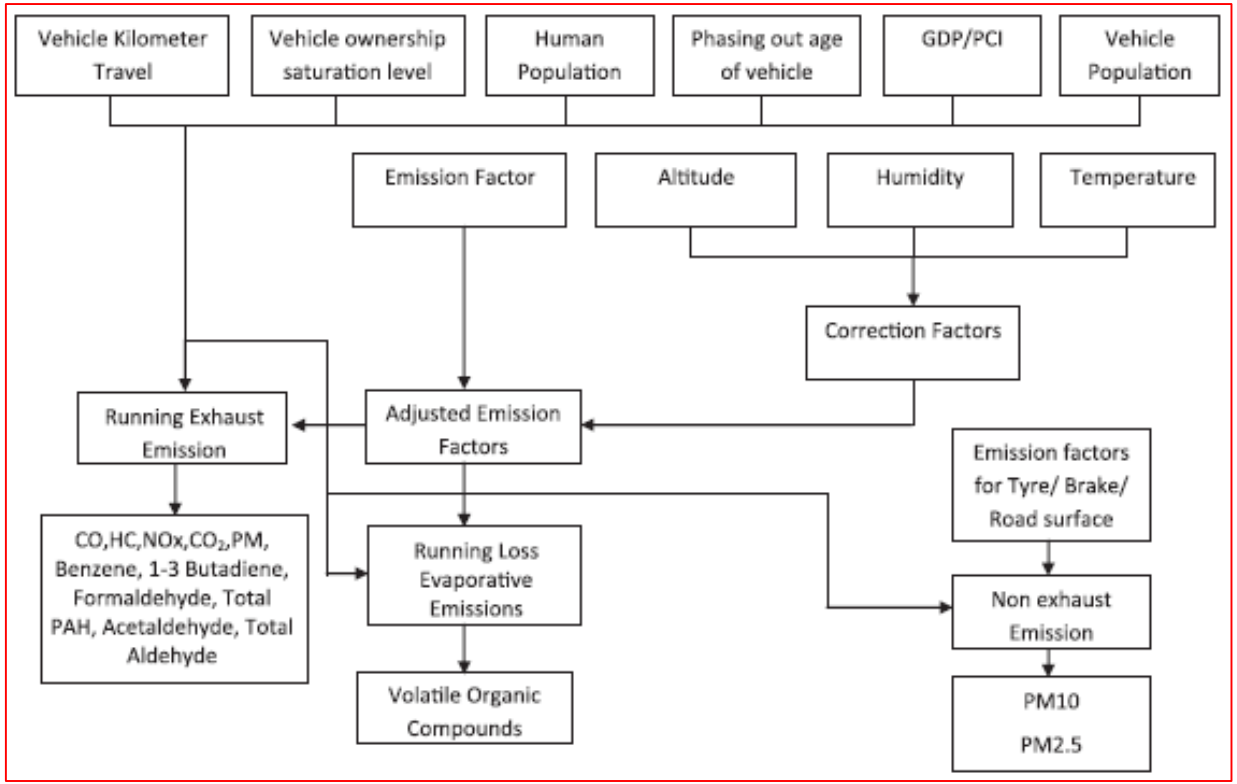
Source: (Ajay Singh Nagpure, 2011)

Assumptions in the VAPI model

- The growth in vehicles per 1000 persons over time is assumed to follow a Sigmoidal (S-shaped) relationship.
- One car plus two wheelers per family are assumed to be the Saturation level.
- The seating capacities of auto-rickshaws and taxis are 3 and 5 respectively, implying that the saturation levels per 1000 persons are assumed to be 330 and 200 respectively.

Source: (Ajay Singh Nagpure, 2011)

Core Architecture of the VAPI model



Source: (Ajay Singh Nagpure, 2011)

Pollutants considered in VAPI model

- Running Exhaust emissions
  - CO, HC, NO_X, CO_2, PM, Benzene, 1-3 Butadiene, Formaldehyde, Total PAH, Acetaldehyde, Total Aldehyde
- Running loss Evaporative emissions
- Non exhaust emissions
  - Emissions from Brakes and Tyres
    - PM_{10} and $PM_{2.5}$

Source: (Ajay Singh Nagpure, 2011)

Running Exhaust emissions

$$EFa, p, i = EF_{p,i} \times CT_{,p,i} \times C_{H,p,i} \times C_{A,p,i}$$

$$E_{t,p,i} = Pi \times EFa_{,p,i} \times V_i \times D_{t,i}$$

The **main concept** in calculating the running exhaust emissions in the VAPI model is to apply an emission factor with climatic and geographical correction factors.

Equation 1

Equation 2

Where,

$D_{t,i}$ = Annual travelling days of i vehicle category

$E_{a,p,i}$ = Adjusted emission rate of pollutant p from i vehicle category

$EF_{p,i}$ = Emission factor of pollutant p from i category (g or mg km^{-1})

$E_{p,i}$ = Total emissions of pollutant p from i vehicle category (unit depends on EF unit)

$C_{t,p,i}$ = Temperature correction factor of pollutant p of vehicle i category

$C_{H,p,i}$ = Humidity correction factor of pollutant p of vehicle i category

$C_{A,p,i}$ = Altitude correction factor of pollutant p of vehicle i category

P_i = On-road Population (model wise) of i vehicle category (numbers)

V_i = Per day distance travel (km day^{-1}) by i vehicle category

Source: (Ajay Singh Nagpure, 2011)

Climatic Correction factors

Temperature Correction factor

$$C_{T.p.i.G} = \left[\left(\frac{C_{p.i.t\text{High}}}{C_{p.i.t\text{Low}}}\right)^{\frac{1}{(t\text{High}-t\text{Low})}} - 1\right] \times 100$$

Equation 3

$$C_{T.p.i} = C_{t\text{Low}} \times (1 + C_{TG})^{(t-t\text{Low})}$$

Equation 4

$C_{T.p.i.G}$ = Growth rate of correction factor according to low and high temperature of pollutant p of i vehicle category
$C_{p.i.t\text{High}}$ = Correction factor in high temperature (t) (40 °C) of pollutant p of i vehicle category
$C_{p.i.t\text{Low}}$ = Correction factor at low temperature (t) (4 °C) of pollutant p of i vehicle category
tHigh = High temperature (t) (40 °C)
tLow = Low temperature (t) (4 °C)
t = Ambient temperature (t in °C)

Humidity Correction factor

$$C_{H.p.i.G} = \left[\left(\frac{C_{p.i.h80\%}}{C_{p.i.h20\%}}\right)^{\frac{1}{(80-20)}} - 1\right] \times 100 \quad (5)$$

Equation 5

$$C_{H.p.i} = C_{h20\%} \times (1 + C_{hG})^{(h-20)} \quad (6)$$

Equation 6

$C_{H.p.i.G}$ = Growth rate of correction factor according to 20% and 80% humidity of pollutant p of i vehicle category
$C_{p.i.h80\%}$ = Correction factor at 80% humidity (h%) of pollutant p of i vehicle category
$C_{p.i.h20\%}$ = Correction factor at 20% humidity (h%) of pollutant p of i vehicle category
h = Ambient humidity (%)

Source: (Ajay Singh Nagpure, 2011)

Geographical Correction factors

Altitude Correction factor

$$C_{A\,p\,i\,G} = \left[\left(\frac{C_{p\,i\,a1700\ \mathrm{m}}}{C_{p\,i\,a950\ \mathrm{m}}} \right)^{\frac{1}{(1700-950)}} - 1 \right] \times 100$$

Equation 7

$$C_{A\,p\,i} = C_{a950\ \mathrm{m}} \times (1 + C_{AG})^{(a-950)}$$

Equation 8

$C_{A\,p\,i\,G}$ = Growth rate of correction factor according to 950 m and 1700 m of pollutant p of i vehicle category

$C_{p\,i\,a1700\ \mathrm{m}}$ = Correction factor at 1700 m altitude (m) of pollutant p of i vehicle category

$C_{p\,i\,a950}$ = Correction factor at 950 m altitude (m) of pollutant p of i vehicle category

a = altitude (m)

Source: (Ajay Singh Nagpure, 2011)

Vehicle Characteristics

On-road Vehicle population, vehicle projections and phasing out of vehicles included in VAPI model.

Vehicle population

- Present and past registered vehicles are considered in the VAPI model

The onroad vehicle population for a year is found out by projecting the vehicle population for the past year (Eq. 9 & 10) and then subtracting from the cumulative registered vehicle population upto that year (Eq. 11)

$$X_{Pop.i} = \left[\left(\frac{RPop.i_{Final}}{RPop.i_{Initial}}\right)^{\frac{1}{y}} - 1\right] \times 100$$

Equation 9

$X_{Pop.i}$: Average geometric rate of annual growth of population (Pop) of i vehicle category
$RPop.i_{Initial}$: Registered population of i vehicle category at the initial year of the period
$RPop.i_{Final}$: Registered population of i vehicle category at the final year of the period
y: Length of time (years) between the initial year and final

$$RPop.i_{Final} = RPop.i_{Initial} \times (1 + x_{Pop.i})^{y}$$

Equation 10

$$Pop_i = RPop._{C\bar{i}} - RPop._{(C-Z)i}$$

Equation 11

Pop_i = Total on-road population of i vehicle category.
$RPop._{C\bar{i}}$ = Registered population of i vehicle category at current year C.
$RPop._{(C-Z)i}$ = Registered population of i vehicle category, in current year minus phasing out age of vehicle where Z is the phasing out age of vehicle.

Source: (Ajay Singh Nagpure, 2011)

Vehicle Characteristics (cont'd..)

Vehicle population projection

- Vehicle population is based on the GDP and per capita income.
- The Gompertz model is used, in the VAPI model.

Gompertz model

Equation 12

$$V_{pt} = \lambda e^{\alpha e^{\beta G}}$$

Where, λ is the saturation level (measured in vehicles per 1000 people), G is GDP or per capita income and α (slope) and β (Intercept) are parameters, which define the shape or curvature of the function. The parameter values α and β are calculated

Equation 13

$$\alpha = \frac{n(\sum GP_{it}) - (\sum G)(\sum P_{it})}{n(\sum (G^2) - (\sum G)^2}$$

Equation 14

$$\beta = \frac{(\sum P_{it})(\sum (G^2)) - (\sum x)(\sum GP_{it})}{n(\sum (G^2) - (\sum G)^2}$$

Where

G = GDP for commercial and public transport and PCI for private vehicle

P_{it} = Per thousand vehicle population of i category of vehicle in particular year (t)

n = years for calculating

Source: (Ajay Singh Nagpure, 2011)

Vehicle model wise Calculation

Base emission factors are based on vehicle age and model year

Registered vehicle population in a year

$$RPop._{m,i} = RPop._{ci} - RPop._{(c-1)i}$$ Equation 15

Where

$RPop._{mi}$ = Each year newly registered (model wise) population of i category vehicle

$RPop._{ci}$ = Population of i category vehicle in current year c

$RPop(c\text{-}1)i$ = Population of i category vehicle in previous (current year-1) years

Percentage of vehicle model present in a year

$$\%RPop._{m,i} = \frac{RPop._{m,i} \times 100}{RP_{Ci}}$$ Equation 16

$\%\ Rpop._{mi}$ = Vintage/Model year wise percentage of population of i category vehicle in each year.

After getting percentage of model wise vehicle population for registered vehicle population in each year this percentage is applied for on-road vehicles

$$Pop_{m,i} = \frac{Pop_i * \%RPop._{m,i}}{100}$$ Equation 17

Source: (Ajay Singh Nagpure, 2011)

Running Evaporative missions

- Running losses from petrol vehicles are largest compared to other sources.
  - Ex. Diurnal emissions, hot soak emissions and resting losses.

- VAPI model calculates the Evaporative adjusted emission factor from the equation:

$$E_{a,p,i} = EFp_{,i} \times CT_{,p,i} \times C_{A,p,i}$$

Equation 18

Vehicular evaporative emissions are the largest contributors to the total evaporative emissions of VOCs.

Running losses are the result of vapor generated in gasoline tanks during vehicle operations.

Source: (Ajay Singh Nagpure, 2011)

Non Exhaust emissions

- Non-exhaust PM from road traffic is generated mechanically by abrasion (wear) of tyre, brake, and road pavement and by the resuspension process of road dust

- VAPI model calculates the Non exhaust emissions using the equation:

$$E_{tpi} = Pi \times Ep_i \times V_i \times D_i$$

Equation 19

Where,

D_i = Annual travelling days of I vehicle category in year

EF_{pi} = Emissions factor of pollutant p from I vehicle category (g or mg km^{-1})

E_{pi} = Total emissions of pollutant p from I vehicle category (unit depends on EF unit)

P_i = Pollution from I vehicle category

V_i = Per day distance travel (km day^{-1}) by I vehicle category

Source: (Ajay Singh Nagpure, 2011)

Conclusion

- Vehicular emission models are mainly developed by various countries such as US, Europe and Australia.

- These complex models are able to predict and evaluate the emissions from vehicles of all categories involving various aspects such as technology, infrastructure and vehicle driving cycles.

- However, most of these models are country specific and processes on the extensive native data sets available in those developed nations.

- The VAPI model is a simple vehicular emission model designed for developing countries, like India.

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Thank You